

Pantheon+ Unbinned Likelihood and Planck Distance Priors: Two Long-Deferred Validations of the C1 Locked Modulation

LVC v14 — Lagrangian Variable Cosmology, Version 14

LUMENPIXEL (independent researcher)

Computational assistance: Claude (Anthropic)

May 2026

Abstract

Versions 10 through 13 of this work consistently identified two limitations that were not addressed in the corresponding numerical analyses: the use of a 13-bin Pantheon+ supernova sample built from a fiducial cosmology rather than the official 1701-SN unbinned likelihood, and the use of only the Planck angular acoustic scale θ_* rather than fuller compressed Planck information. In v14 we address both, holding the C1 model fixed at the v12 family $n = 1$ lock $z_c = 1/\sqrt{3} + 1/(3\pi)$, $w = 2/(3\pi)$.

Replacing the binned Pantheon+ with the official 1701-SN sample and full STAT + SYS covariance from Brout et al. (2022), and keeping the rest of the v13 baseline dataset, gives $\Delta\text{BIC} = -18.64$ for C1 versus ΛCDM at $N = 1724$. The v13 §3.5 worry that the result depended on the fiducial Ω_m used in binning is therefore removed by construction: the unbinned likelihood is fiducial-free, and the C1 advantage is *stronger*, not weaker, than the v13 baseline value of $\Delta\text{BIC} = -11.57$.

Adding the Planck 2018 compressed distance priors (R, ℓ_A, ω_b) from Chen, Huang & Wang (2019, TT,TE,EE+lowE+lensing) to the likelihood — making this a global late-time + compressed-CMB analysis — gives $\Delta\text{BIC} = -11.38$ at $N = 1727$. The result remains in the “very strong” regime by the standard Jeffreys’ scale. The v13 §3.15 concern, that forcing $r_d = 147$ Mpc would eliminate the C1 advantage, is not realised here: the Planck distance priors pull the derived r_d from ~ 137 to ~ 145 Mpc but do not eliminate the C1 preference. Furthermore, C1 fits the Planck distance priors *slightly better* than ΛCDM does (calibrated χ^2_{DP} : 15.03 vs 20.58). The Hubble parameter preferred by C1 shifts from 73.4 (Task 1, no DP) to 69.4 (Task 2, with DP), partially absorbing the SH0ES/Planck Hubble tension at the cost of a $\sim 12\sigma$ contribution to χ^2 from the SH0ES residual.

We do not claim resolution of all the open problems identified in v13 §6. Two limitations carried forward unchanged: a full Boltzmann-level Planck likelihood (plik TT,TE,EE high- ℓ) is not used in this work, and the model still has no first-principles Lagrangian formulation. We also note a $\sim 0.5\%$ systematic offset in our compressed-Planck ℓ_A computation versus full Boltzmann, calibrated out of the absolute χ^2 values; this calibration applies identically to ΛCDM and C1 because the sound-horizon integrand is unaffected by $T(z)$ at $z \gg 1$, so the relative comparison (ΔBIC) is unaffected.

1 Setup

The C1 model is unchanged from v12/v13:

$$T(z) = 1 + A \cdot \exp\left[-\left(\frac{z - z_c}{w}\right)^2\right], \quad z_c = \frac{1}{\sqrt{3}} + \frac{1}{3\pi}, \quad w = \frac{2}{3\pi}, \quad (1)$$

with $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$. The locked values $z_c \approx 0.6834$ and $w \approx 0.2122$ correspond to the family $n = 1$ rung defined in v12. We carry only C1 forward to v14; C2 (PowRat) is not analysed here.

The data sets used in this paper are the v13 BAO + CMB-acoustic + local H_0 baseline, with the supernova block replaced as described in §2, and (for Task 2) the Planck compressed distance priors added as described in §3:

- **Pantheon+ unbinned:** 1701 SN with full STAT + SYS covariance, including 77 Cepheid-host calibrators (Brout et al. 2022; Scolnic et al. 2022). Loaded from Pantheon+SH0ES.dat and Pantheon+SH0ES_STAT+SYS.cov.
- **DESI DR2 BAO:** 1 isotropic D_V at $z = 0.295$ and six anisotropic D_M/r_d , D_H/r_d pairs at $z = 0.510, 0.706, 0.934, 1.321, 1.484, 2.330$ (DESI Collaboration 2025). 13 measurements.
- **BOSS DR12** D_M, D_H at $z = 0.38$. 2 measurements.
- **eBOSS DR16:** D_M, D_H at $z = 0.698, 1.480, 2.334$ and D_V at $z = 0.845$. 7 measurements.
- **Planck** θ_* : $\theta_* = 0.010409$, used to derive r_d in geometric fits.
- **SH0ES** H_0 : $H_0 = 73.04 \pm 1.04$ km/s/Mpc (Riess et al. 2022).
- **Planck distance priors** (Task 2 only): $R = 1.7493 \pm 0.0049$, $\ell_A = 301.462 \pm 0.090$, $\omega_b = 0.02236 \pm 0.00015$, with the correlation matrix from Chen, Huang & Wang (2019), Table 1 (TT,TE,EE+lowE+lensing).

The v13 baseline used 13 binned Pantheon+ points and 22 BAO + 1 θ_* + 1 H_0 measurements for $N = 37$. Task 1 of v14 has $N = 1724$ (1701 + 22 + 1; θ_* is consumed by the derivation of r_d). Task 2 of v14 has $N = 1727$ (Task 1 plus three distance-prior measurements).

2 Task 1: Pantheon+ unbinned likelihood

2.1 Likelihood

Following Brout et al. (2022), for each SN i we form the residual

$$d_i = m_{B,i} - \mu_{\text{pred},i} - M_B, \quad (2)$$

where $m_{B,i}$ is the Tripp-corrected apparent magnitude (`m_b_corr` in the public file), M_B is the standard absolute magnitude (a single common nuisance parameter), and the predicted distance modulus is

$$\mu_{\text{pred},i} = \begin{cases} \mu_{\text{Cepheid},i} & \text{if SN } i \text{ is in a Cepheid host,} \\ 5 \log_{10}[(1 + z_{\text{Hel},i}) D_C(z_{\text{HD},i})] + 25 & \text{otherwise.} \end{cases} \quad (3)$$

The chi-squared is $\chi_{PP}^2 = (d - M_B \mathbf{1})^\top C^{-1} (d - M_B \mathbf{1})$, with C the full 1701×1701 STAT+SYS covariance matrix. We profile M_B analytically: $\widehat{M_B} = (\mathbf{1}^\top C^{-1} d) / (\mathbf{1}^\top C^{-1} \mathbf{1})$.

2.2 Sanity check

At $H_0 = 73.04$ km/s/Mpc and $\Omega_m = 0.30$ (close to the joint best fit), we obtain $\chi_{PP}^2 = 1526.49$, i.e. $\chi^2/\text{d.o.f} = 0.898$, in good agreement with the value ~ 0.92 reported by Brout et al. (2022). The profiled $\widehat{M_B} = -19.27$ matches the standard SH0ES-anchored value to within 0.02 mag.

2.3 Results

We jointly fit Λ CDM ($H_0, \Omega_m, k = 2$) and C1 ($H_0, \Omega_m, A, k = 3$), with r_d derived in both cases via $r_d = \theta_* D_M(z_*)$ where $D_M(z_*)$ is computed including radiation ($\omega_y h^2 = 2.473 \times 10^{-5}$, $N_{\text{eff}} = 3.046$). The full Task 1 likelihood is

$$\chi_{\text{Task 1}}^2 = \chi_{PP}^2 + \chi_{\text{BAO}}^2 + \frac{(H_0 - 73.04)^2}{1.04^2}. \quad (4)$$

Table 1: Task 1 best fits. $N = 1724$.

Model	k	χ^2	H_0 [km/s/Mpc]	Ω_m	A	r_d [Mpc]
Λ CDM	2	1565.835	73.528	0.2773	—	139.27
C1	3	1539.748	73.411	0.2917	+0.0557	136.51
$\Delta\chi^2$ (C1 – Λ CDM)		–26.087				
ΔBIC (C1 – Λ CDM)		–18.635				

The C1 advantage in Task 1 is $\Delta\text{BIC} = -18.64$, larger in magnitude than the v13 baseline value of -11.57 . The Pantheon+ chi-squared contribution decreases from 1531 to 1525 in C1; most of the improvement comes from the BAO block (where C1 reduces χ^2_{BAO} from ~ 27 to ~ 14).

2.4 Resolution of the v13 §3.5 worry

In v13 §3.5 we documented that the binned-Pantheon+ analysis depends on the fiducial Ω_m used to construct the bins, with ΔBIC ranging from -0.45 at $\Omega_m^{\text{fid}} = 0.27$ to -4.36 at $\Omega_m^{\text{fid}} = 0.334$ when the binning was redone. Task 1 of v14 removes this dependence by construction: the unbinned likelihood treats each of the 1701 SN as an independent measurement with no prior assumption on cosmology. The fact that $\Delta\text{BIC} = -18.64$ is on the strong end of the v13 fiducial-binning range, rather than the weak end, is consistent with v13’s binning at $\Omega_m^{\text{fid}} = 0.334$ being the most representative of the data, but the conclusion is now independent of any binning choice.

3 Task 2: Planck compressed distance priors

3.1 Distance priors and their information content

The Planck CMB primary anisotropies, when used to constrain late-time modified $H(z)$ models like ours, are well approximated by three compressed parameters (Chen, Huang & Wang 2019):

$$R = \sqrt{\Omega_m H_0^2} D_M(z_*)/c, \quad \ell_A = \pi D_M(z_*)/r_s(z_*), \quad \omega_b = \Omega_b h^2. \quad (5)$$

R depends on the late-time integral of $1/H(z)$, hence on $T(z)$ directly. ℓ_A depends on the ratio of that integral to the sound horizon at decoupling, which is integrated from z_* to high redshift in a regime where $T(z) \rightarrow 1$ by construction. ω_b is unaffected by $T(z)$.

For Λ CDM at the Planck TT,TE,EE+lowE+lensing best fit ($H_0 = 67.36$, $\Omega_m = 0.3153$, $\omega_b = 0.02237$), the values of (R, ℓ_A, ω_b) should match the Chen+19 reported means. Using a direct numerical integration with the Hu & Sugiyama (1996) z_* formula, we find a -0.018% offset in R and -0.478% offset in ℓ_A . Both offsets are caused by the well-known limitation of fitting formulas relative to a full Boltzmann calculation (CAMB or CLASS); they are independent of which late-time model we are fitting because the high-redshift sound-horizon integral is unaffected by $T(z)$. We absorb the offsets via a multiplicative calibration $\text{CAL}_R = 0.999818$ and $\text{CAL}_{\ell_A} = 0.995218$ applied to both Λ CDM and C1 alike, restoring exact agreement with the Planck fiducial. This calibration cancels in $\Delta\chi^2$ and ΔBIC .

3.2 Results

We add the distance-prior likelihood $\chi^2_{DP} = (\mathbf{x} - \mathbf{x}_{\text{Planck}})^T C_{DP}^{-1} (\mathbf{x} - \mathbf{x}_{\text{Planck}})$ with $\mathbf{x} = (R, \ell_A, \omega_b)$ to the Task 1 likelihood, and refit. Λ CDM now carries ω_b as a third free parameter ($k = 3$), and C1 carries it as a fourth ($k = 4$).

Table 2: Task 2 best fits. $N = 1727$.

Model	k	χ^2	H_0	Ω_m	ω_b	A	r_d [Mpc]
Λ CDM	3	1606.94	70.31	0.2775	0.02285	—	145.55
C1	4	1588.10	69.43	0.2865	0.02279	+0.0442	145.34
$\Delta\chi^2$		−18.83					
ΔBIC		−11.38					

The chi-squared decomposition is informative:

Table 3: Task 2 χ^2 decomposition by data block.

Block	Λ CDM	C1
Pantheon+ (1701 SN)	1545.99	1545.06
DESI DR2 + BOSS + eBOSS BAO (22)	33.48	15.95
SH0ES H_0	6.89	12.06
Planck distance priors (3)	20.58	15.03
Total	1606.94	1588.10

Three observations are worth recording.

1. Most of the C1 improvement still comes from the BAO block, which is also where the v13 baseline locates it: $\Delta\chi^2_{\text{BAO}} \approx -17.5$.
2. C1’s SH0ES χ^2 is *larger* than Λ CDM’s (12.1 vs 6.9). This reflects that C1’s preferred $H_0 = 69.4$ now sits about 3.5σ below the SH0ES central value, while Λ CDM at 70.3 sits about 2.6σ below. C1 absorbs more of the Hubble tension by lowering its preferred H_0 when the Planck DP forces r_d closer to 147 Mpc.
3. C1’s distance-prior χ^2 is *lower* than Λ CDM’s (15.03 vs 20.58). C1 fits the compressed Planck data slightly better than Λ CDM does, even after the calibration is applied identically to both. This is consistent with C1 helping reconcile the late-time and early-time constraints, although the absolute $\chi^2_{\text{DP}} \sim 15$ for C1 is not small (the three-DP measurement system is in $\sim 3.4\sigma$ tension with the late-time data even for C1).

3.3 Resolution status of the v13 §3.15 concern

In v13 §3.15 we forced $r_d = 147.05$ Mpc and found that C1’s ΔBIC collapsed from -11.57 to -3.49 . This was identified as a serious concern: if BBN-compatible r_d is required, the C1 advantage deteriorates by $+8.08$. Task 2 is the rigorous version of that test: instead of forcing r_d , we let the Planck distance priors constrain the early-universe physics that determines r_d .

The result is that r_d is pulled from ~ 137 Mpc (Task 1) to ~ 145 Mpc (Task 2), a substantial shift, but it does not reach the nominal Planck value of 147 Mpc. Both Λ CDM and C1 settle at very similar $r_d \approx 145$ Mpc; the late-time data prefer r_d slightly below the Planck value, consistent with the well-known SH0ES/Planck Hubble tension. The C1 advantage is reduced from Task 1’s -18.64 to Task 2’s -11.38 , but remains in the “very strong” regime by the Jeffreys’ scale.

A natural reading: the v13 §3.15 “ $r_d = 147$ forced” test was a worst-case scenario, in which all the DP information was distilled into a single rigid constraint on r_d alone. In the more realistic Task 2 analysis, where Planck information is a soft Gaussian prior on

(R, ℓ_A, ω_b) rather than a hard constraint on r_d , the tension does not eliminate the C1 advantage.

4 Combined view and remaining limitations

Table 4: Comparison ladder: v13 baseline and v14 results.

Analysis	N	SN treatment	ΔBIC
v13 baseline	37	13-bin (fid. $\Omega_m = 0.334$)	−11.57
v14 Task 1 (PP unbinned, no DP)	1724	full 1701-SN + STAT+SYS cov.	−18.64
v14 Task 2 (PP unbinned + Planck DP)	1727	full 1701-SN + STAT+SYS cov.	−11.38

The conclusion is uniform across all three analyses: at the C1 lock $z_c = 1/\sqrt{3} + 1/(3\pi)$, $w = 2/(3\pi)$, the model is preferred over ΛCDM by Bayesian information criteria at the “very strong” level. The strength varies by a factor of ~ 1.6 depending on which external constraints are folded in, but the qualitative result does not.

Limitations not addressed by v14. We list these explicitly so they are not discovered for us:

1. **Full Boltzmann Planck likelihood.** Task 2 uses the compressed distance priors, not the full plik high- ℓ TT, TE, EE likelihood. The compressed parameters capture the sensitivity of CMB primary anisotropies to late-time geometry, but they do not test the damping-tail signature of any modification to early-universe physics (e.g., N_{eff} , early dark energy, varying constants) that might be invoked to produce $r_d \approx 145$ Mpc rather than 147 Mpc. A full plik likelihood evaluation is beyond the scope of this work.
2. **Sound-horizon precision.** The $\sim 0.5\%$ offset of our ℓ_A versus full Boltzmann is calibrated out and does not affect ΔBIC , but it leaves the absolute χ^2_{DP} values uncertain at that level.
3. **Lagrangian formulation.** The C1 modulation $T(z)$ remains a phenomenological multiplicative factor on $H(z)$. There is no derived action principle, no perturbation-level consistency proof, and the joint $f\sigma_8$ analysis (v11 §3, v13 §3.6) still uses a scale-independent linear-growth ODE that assumes no modification of the gravitational constant. The model name “Lagrangian Variable Cosmology” remains aspirational rather than actual.
4. **Family hypothesis.** The v12/v13 family $z_c(n) = 1/\sqrt{3} + n/(3\pi)$, $w(n) = (n+1)/(3\pi)$ remains tested only at $n = 0$ (DR1) and $n = 1$ (DR2). DESI DR3 will be the first real test of whether the family is a meaningful structure.
5. **C2 not analyzed.** The C2 (PowRat $w = 2/\pi$) candidate identified in v13 is not carried through v14. v13 §3.15 already documented that C2 was nearly eliminated by the BBN constraint, and this remains expected to hold under the more realistic Task 2.

Net status. Two of the four limitations identified in v13 §6 (“binned SN” and “no full Planck”) are partly addressed: the SN treatment is now full unbinned, and the CMB enters through compressed distance priors. The remaining two (Lagrangian formulation, family hypothesis) are unchanged. We characterise this v14 as a removal of two specific data-treatment concerns from v10–v13, not as a settled case for the C1 model.

5 Reproducibility

All numerical results are reproduced by `lvc_v14_reproduce.py`, which extends `lvc_v13_reproduce.py` with: (i) the Pantheon+ unbinned likelihood, including matrix inversion of the 1701×1701 STAT+SYS covariance and analytic profiling of M_B , and (ii) the Planck distance-prior likelihood, with explicit calibration to remove the fitting-formula offset at the Planck fiducial.

External data used:

- Pantheon+ data and covariance: `github.com/PantheonPlusSH0ES/DataRelease`, files `Pantheon+SH0ES.dat` and `Pantheon+SH0ES_STAT+SYS.cov` from the `4_DISTANCES_AND_COVAR` subdirectory.
- Planck distance priors: Chen, Huang & Wang (2019), ApJ 880:65, arXiv:1808.05724, Table 1, “TT,TE,EE+lowE+lensing”.

Tabulated χ^2 values are reproduced within ≤ 0.1 and BIC values within ≤ 0.2 . Computational platform: Python 3, `scipy differential_evolution` (multi-seed) followed by `minimize(method='Nelder-Mead')`. Total runtime to reproduce the full v14 paper from scratch is approximately 4 minutes on a single CPU core.

6 Note on scope

This v14 paper reports a specific data-treatment update to v10—v13. It does not introduce new model forms, does not relax the C1 lock, and does not claim a derivation. The v13 stress-test survey is not superseded; it is supplemented. v13 and v14 together constitute the current numerical position of the LVC program, with v13 documenting the model’s behavior under structural perturbations and v14 documenting its behavior under upgraded data.